

# Noppayut Sriwatanasakdi

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## EXPERIENCE

### Square Enix AI & Arts Alchemy Co., Ltd.

Sep 2022 – Present

#### Natural Language Processing Engineer

Tokyo, Japan

Develop a real-time conversational AI framework and virtual avatar development tools. Collaborate closely with game designers and programmers to translate feature requirements into technical implementations.

- Lead the design and development of machine learning-based components such as content moderation, user intent and entity recognition, and text style transformation. Responsible for the entire development cycle, including research, data preparation, model training/fine-tuning, evaluation, model optimization, and integration into the conversational framework.
- Provide practical guidance on effective use of conversational AI and large language models (LLMs), including prompt engineering best practices and safety strategies for conversation scenarios.
- Lead the development of an internal MLOps web application that focuses on data management and experimentation tracking. In charge of project management, direction setting, system design, and back-end development. The tool allows non-technical team members to independently manage training datasets and experiment with ML models. Drastically reduced the model iteration cycle from days to hours with minimal engineering support.
- Design and implement systems to ensure that avatar responses are safe and stay within permitted topics during controversial scenarios. Responsible for developing guardrail applications and red-teaming workflows.
- Developed automated evaluation metrics for assessing stylistic text generation quality. The approach reduced manual evaluation time while maintaining consistency and objectivity.
- **Tools:** C++, Python, PyTorch, Transformers (HuggingFace), TRL (Transformer Reinforcement Learning), OpenAI, Milvus, vLLM, Triton inference server, gRPC, ONNX, Docker, Google Cloud Platform, FastAPI, Pydantic, PostgreSQL, Redis

### Asurion Japan Holdings G.K.

Nov 2019 – Aug 2022

#### Data scientist

Tokyo, Japan

Developed AI/data solutions to support smartphone protection services of major Japanese carriers. Provided technical consultation, presented demos and performance estimates to business users. Collaborated with DevOps to deliver services. Led development and/or maintenance of several projects:

- **Troubleshoot Chatbot:** A chatbot that solves user's smartphone issues by showing relevant FAQ articles. Achieved 10% gain in absolute trouble resolution rate and 25% in answer display rate over the previous system. Served 30k users per month in average.
- **Keyword suggestion:** attached to a search box, this system helps users find right keywords to reach desirable FAQ articles. The system can lead users to the solution articles roughly 80% of time on queries that can be resolved. Receive 1,000 queries/day.
- **Webclaim validator:** a system for validating user input when filing a device claim through web channel. Significantly reduce the cost of using human validators. Serving 30-40 claims per day.
- **Telemetry trouble prediction:** a method to predict smartphone issues using real-time telemetry data (e.g. app usage history). The method is more accurate than guessing with most-common-trouble baselines in top 3, 4, and 5 predictions by large margins.
- Other projects such as identification of topic boundaries in conversations, customer behavior analysis, user comment classification and development of text analytic tools for non-technical users.
- **Tools:** Scikit-learn, XGBoost, Gensim, PyTorch, Torchtext, Transformers, Matplotlib, Seaborn, MeCab, Pandas, Docker, Flask, FastAPI, AWS EC2, Lambda

### Works Applications Co., Ltd.

Apr 2018 – Oct 2019

#### R&D Engineer

Tokyo, Japan

- Worked on research and development of machine learning and NLP solution to business problems such as FAQ chatbot, facility management, resignation prediction, etc.
- **Tools:** Pandas, PyTorch, Seaborn, Scikit-learn, Google Cloud Platform (GAE), Java, Spring, NLTK, Javascript

## EDUCATION

### Osaka University

Apr 2016 – May 2018

#### Master of Information Science and Technology

- Focused in data mining and machine learning
- Thesis: Classification in Evolving Data streams with Concept Drift Detection

### Chulalongkorn University

Jun 2011 – Jun 2015

#### Bachelor of Engineering in Computer Engineering

- First class honors, GPA: 3.78/4.00

## SKILLS

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**Professional skills** : Machine learning, Natural Language Processing, Data mining, System design, Data visualization

**Languages** : (Preferred) Python, C++, (Experienced) Java, C, TypeScript, Matlab, Scala

**Tools (AI/ML/Viz)** : Scikit-learn, PyTorch, XGBoost, Transformers, ONNX\*, Gensim, Pandas, Numpy, vLLM, Triton, Matplotlib, Seaborn

**Frameworks and Environments** : Git, Jupyter lab/notebook, Linux command line, Docker, Amazon AWS (EC2, Lambda, ECS, etc.), Google Cloud Platform (Compute Engine, Cloud Run, etc.), FastAPI, Flask

**Experienced tools** : MySQL, Redis, PostgreSQL, Spring framework, Selenium

## LANGUAGES

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**Thai** : Native

**English** : Professional working proficiency (TOEIC 960/990 – 2017, TOEFL-iBT 102/120 – 2019)

**Japanese** : Limited working proficiency (JLPT N2 – 2017)

## AWARDS, HONORS, AND EXTRACURRICULAR EXPERIENCES

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**Lecturer and mentor, the 4th and 5th Creative AI Camp by CPALL**

Nov 2021 – Dec 2022

**Winner team of the Thailand NLP Hackathon**

Jul 2020

**Master degree student with Japanese Government Scholarship (Monbukagakusho)**

Apr 2016 – Apr 2018

**Teaching Assistant (TA), Java class, Osaka University**

Oct 2016 – Feb 2017

## PUBLICATIONS

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[1] Chairatanakul, N., **Sriwatanasakdi, N.**, Charoenphakdee, N., Liu, X., and Murata, T. "Cross-lingual Transfer for Text Classification with Dictionary-based Heterogeneous Graph", In: [Findings of EMNLP 2021](#). (Long paper)

[2] **Sriwatanasakdi, N.**, Numao, M., and Fukui, K. "Concept Drift Detection for Graph-structured Classifiers under Scarcity of True Labels", In: [Proc. of ICTAI, 2017](#). (Long paper)

## OPEN SOURCES AND SIDE PROJECTS

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My open source contributions and hobby projects. See more on my [Github](#).

### **NVIDIA/FasterTransformer #460**

- FasterTransformer (FT) is Nvidia's highly optimized transformer models for inference. Fix bugs in a script to convert Huggingface's Japanese GPT2 to FT model.
- Note: FT development is discontinued and transitioned to TensorRT-LLM.

### **PyThaiNLP/pythainlp #723, #735, #751, #752**

- Thai language NLP toolkit. Implemented new features such as keyword extraction and Thai-to-English transliteration. Bug fixing and documentation.